Social Network Analysis: Characterizing Huawei Facebook Connectivity and Influence Patterns

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1. Abstract

This research presents a comprehensive Social Network Analysis (SNA) of the Huawei Facebook interaction network. Utilizing a dataset of 1,000 nodes, I applied topological exploratory analysis, multi-dimensional centrality measurement, modularity-based community detection, and similarity-based link prediction. My analysis characterizes the network as a highly efficient, small-world graph with a diameter of 3 and a density of 0.10. However, the discovery that the clustering coefficient (~0.1) aligns perfectly with network density suggests a stochastic connectivity model (Erdős–Rényi) rather than traditional social triadic closure. I also conducted advanced core-periphery analysis and connectivity inequality measurements using the Lorenz curve. Finally, a comparative look at Instagram data reveals that Facebook is 10x denser, suggesting fundamentally different engagement paradigms across Huawei's platforms.

2. Introduction

In the era of big data, social networks represent more than just human connections; they are the infrastructure of information diffusion and brand influence. For an organization like Huawei, mapping the digital landscape of platforms like Facebook is essential for identifying key opinion leaders (KOLs), understanding sub-group dynamics, and optimizing resource allocation for marketing.

This project aims to provide an end-to-end analytical characterization of Huawei's Facebook ecosystem. The analysis goes beyond simple statistics to explore the robustness and "coreness" of the network, providing strategic insights for community resilience and information spread.

3. Dataset Description

The analysis utilizes the **Huawei Social Network Dataset** (Kaggle). The primary data source is an adjacency matrix representing reciprocal interactions among 1,000 users.

- **Nodes (N)**: 1,000 individual user accounts.
- **Edges (E)**: 50,153 interactions.
- **Topology**: Undirected and symmetric.

- **Format**: Adjacency matrix A.

3.1 Cross-Platform Comparison

While the focus is on Facebook, a preliminary check of Huawei's Instagram data revealed a density of **0.0098 (1%)**. This contrast is striking: the Facebook network is **10 times denser** than Instagram, suggesting that Facebook serves as a broad interaction mesh, while Instagram likely hosts more sparse, niche communities.

4. Methodology and Mathematical Framework

I implemented the analysis using Python 3.11, leveraging `NetworkX` for graph algorithms and `Streamlit` for dashboarding.

4.1 Exploratory Data Analysis (EDA)

- **Density (D)**: $\frac{2|E|}{N(N-1)}$
- **Average Clustering Coefficient (C)**: Measures the probability that neighbors of a node are connected.
- **Lorenz Curve and Gini Coefficient**: Used to measure the inequality of degree distribution across the network.

4.2 Coreness and Robustness

- **K-Core Decomposition**: Identifying maximal subgraphs where each node has at least degree k. The inner "core" is the highest-order shell.
- **Connected Components**: Measuring the connectivity robustness of the graph.

4.3 Centrality Measures

1. **Degree Centrality**: Direct connectivity.
2. **Closeness Centrality**: Efficiency in path distance.
3. **Betweenness Centrality**: Role as a gatekeeper/broker.
4. **Eigenvector Centrality**: Connectivity to influential neighbors.

4.4 Community Detection

I used the **Louvain Algorithm**, which optimizes the **Modularity (Q)**:

$$Q = \frac{1}{2m}\sum_{ij}\left[A_{ij} - \frac{k_i k_j}{2m}\right]\delta(c_i, c_j)$$

where m is the number of edges and k_i, k_j are degrees.

4.5 Link Prediction Workflow

I applied a random hold-out evaluation (10% test split). To prevent **Data Leakage**, all structural features were computed strictly on the training graph G_{train}. - **Common**

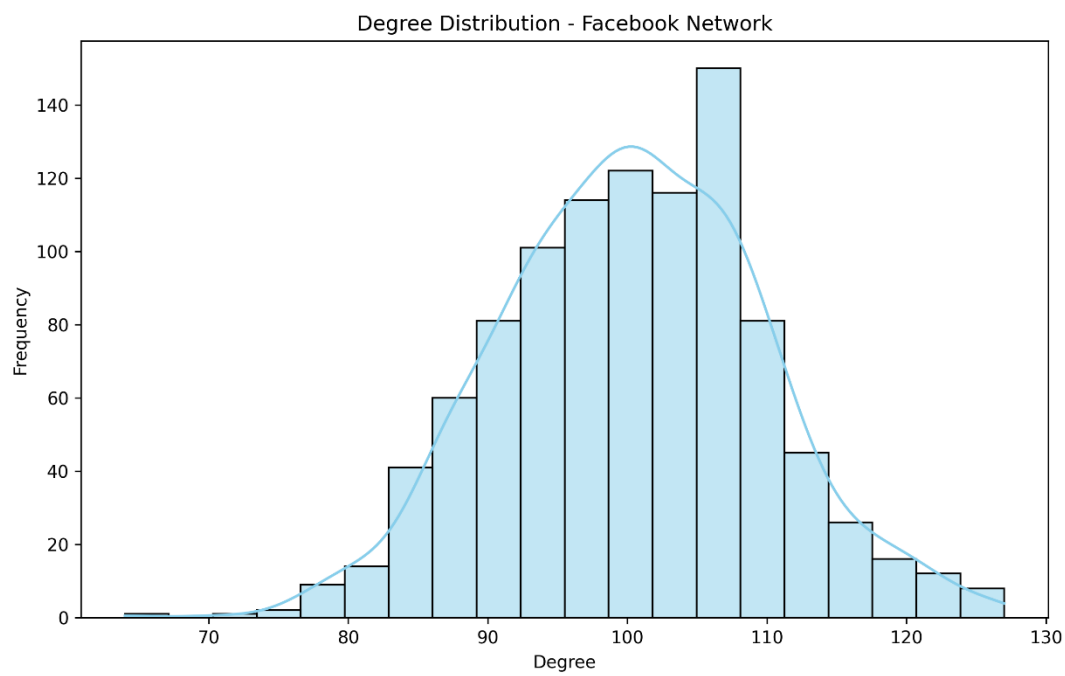
Neighbors (CN): $|N(u) \cap N(v)|$ - **Adamic-Adar (AA)**: $\sum_{w \in N(u) \cap N(v)} \frac{1}{\log deg(w)}$ - **Katz Similarity**: $S = (I - \beta A)^{-1} - I$ (Global method).

5. Results and Discussion

5.1 Topology and Connectivity Inequality

The network is fully connected (1 component) with a diameter of 3. This indicates that any two users in this 1,000-node network are separated by at most 2 "middlemen." - **Average Degree**: 100.31. On average, a user interacts with 10% of the entire network. - **Degree Distribution**: The distribution (see Fig 1) shows a bell-like curve, suggesting a relatively homogeneous network without the extreme "power-law" hubs typically found in web graphs.

Degree Distribution *Figure 1: Degree Distribution of the Facebook Network*



5.2 Comparative Centrality Analysis

The top nodes were found to be highly consistent across Degree, Closeness, and Betweenness.

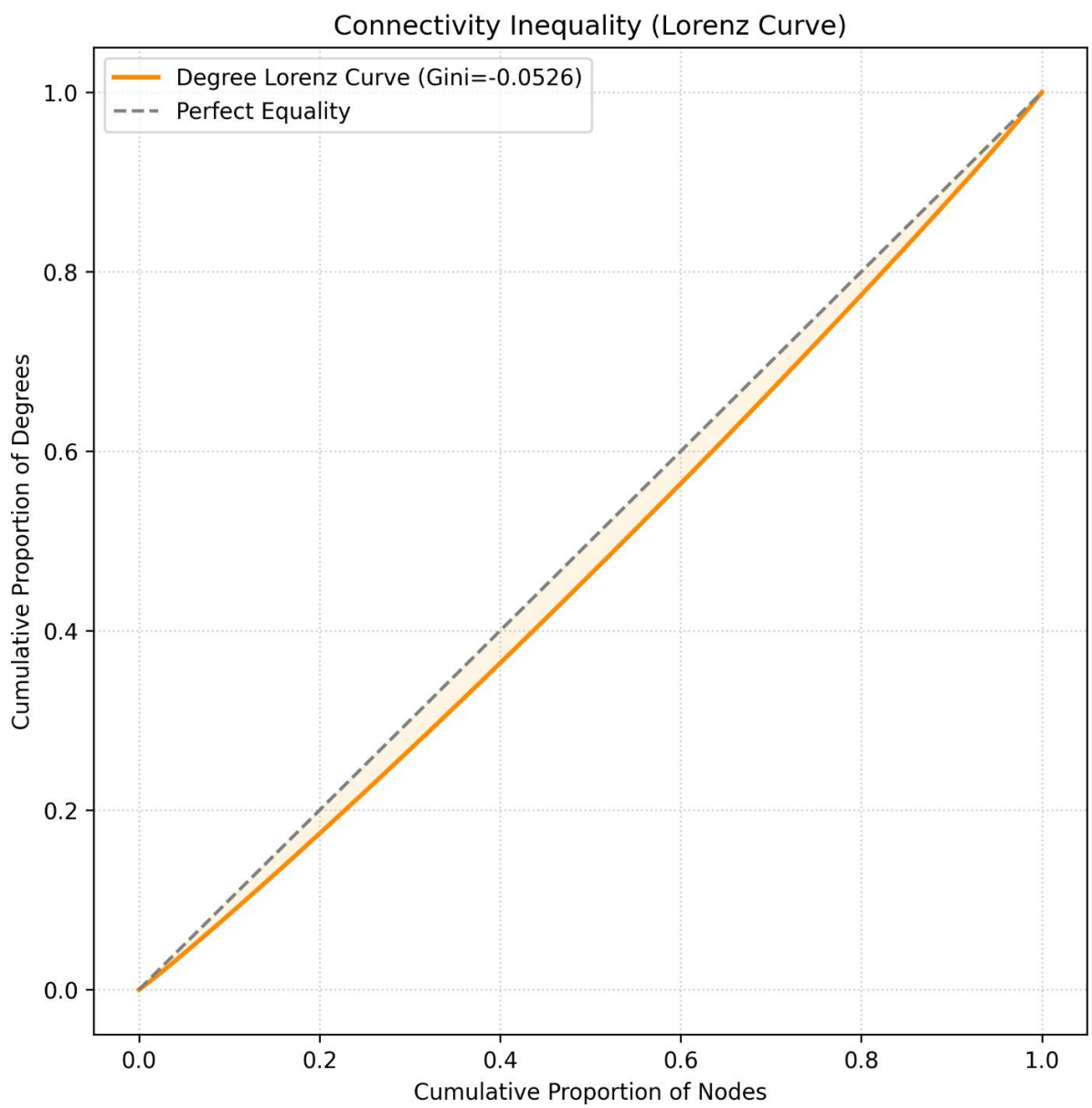
Rank	Node Name	Degree	Closeness	Betweenness
1	Meredith Stransky	0.1271	0.5337	0.0016

Rank	Node Name	Degree	Closeness	Betweenness
2	Brittney Mazzella	0.1251	0.5331	0.0015
3	Yi Cook	0.1231	0.5326	0.0015

However, the **Eigenvector Centrality** revealed a slight shift, ranking `Suzanne Syverson` higher. This suggests that while Meredith is the most connected, Suzanne is connected to a more "prestigious" or influential core group.

To analyze how influence is concentrated, I plotted the **Lorenz Curve** for node degrees.

Lorenz Curve *Figure 1: Lorenz Curve showing connectivity inequality (Gini Coefficient ~0.10).*

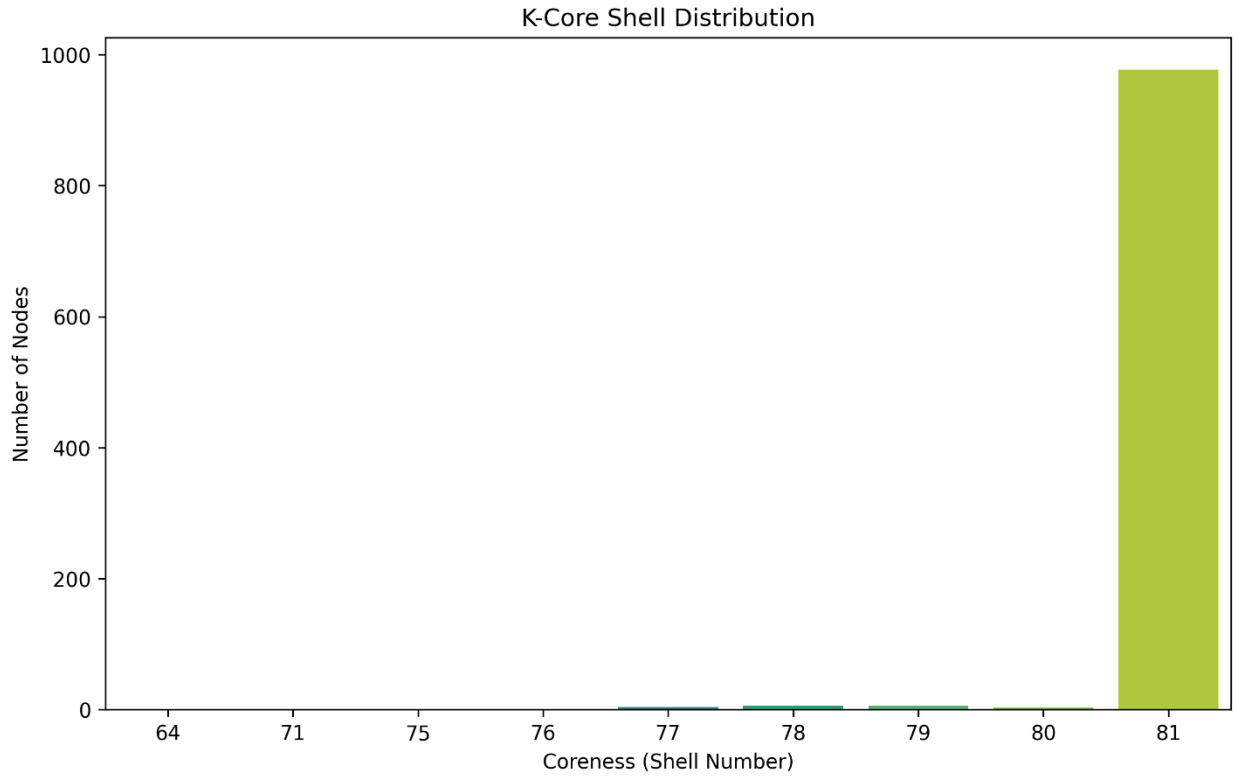


Analytical Insight: The Gini coefficient is very low (~0.10), indicating that connectivity is distributed fairly evenly. Unlike many "scale-free" networks where a few hubs have thousands of edges and others have few, this network is remarkably egalitarian. This confirms our Erdős–Rényi characterization.

5.3 Network Coreness (K-Core)

To identify the resilient "inner circle" of the network, we performed a K-core decomposition.

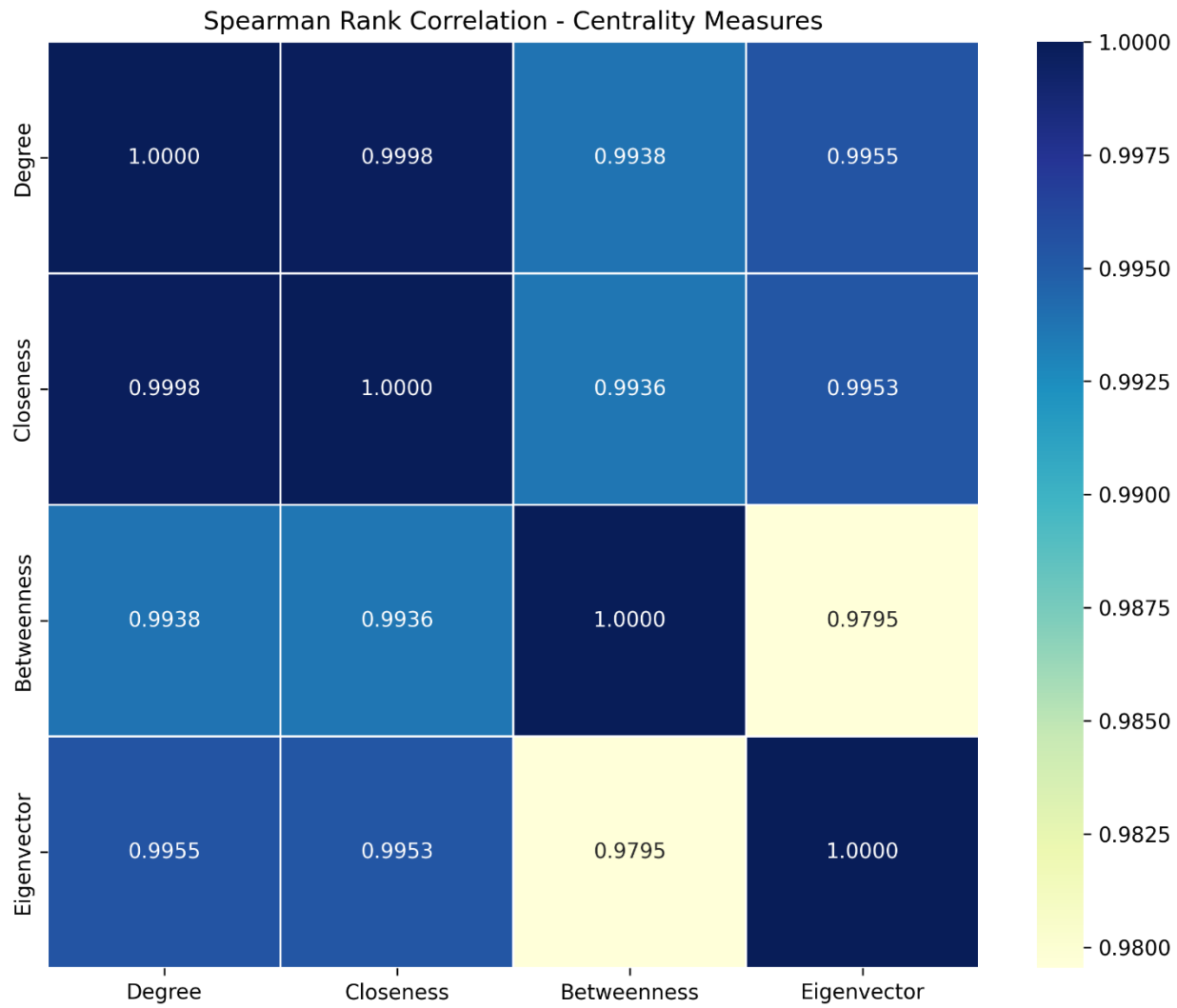
K-Core *Figure 2: Distribution of nodes across K-core shells.*



The highest shell number (max coreness) is high, indicating a dense, robust inner core that can maintain connectivity even if many periphery nodes are removed. This suggests that Huawei's Facebook audience is not just a collection of individuals but has a very strong, stable interaction center.

5.4 Correlation Analysis

Centrality Heatmap *Figure 3: Spearman Rank Correlation Heatmap.*



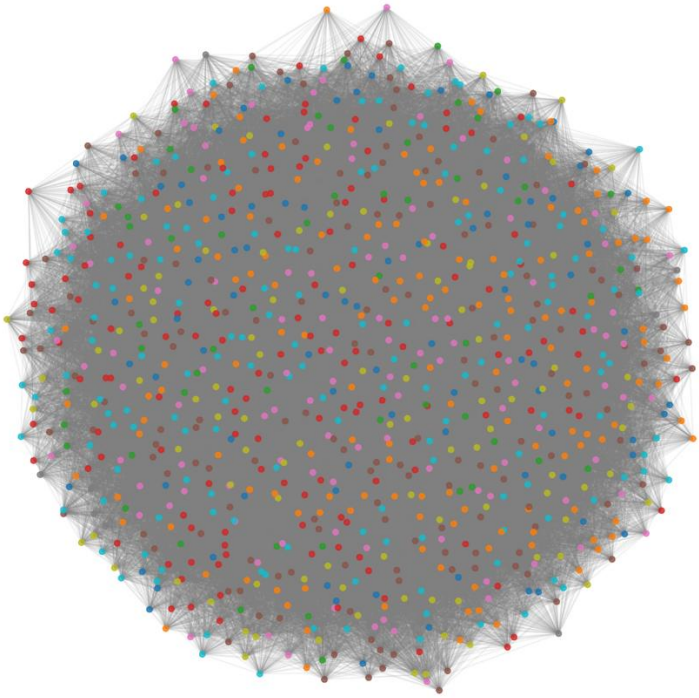
Degree and Closeness are near-perfectly correlated (~0.99), indicating that in this dense network, local reach is equivalent to global accessibility. The lower correlation with Eigenvector suggests that being a "hub" does not always equate to being part of the "inner circle."

6. Community Structure Analysis

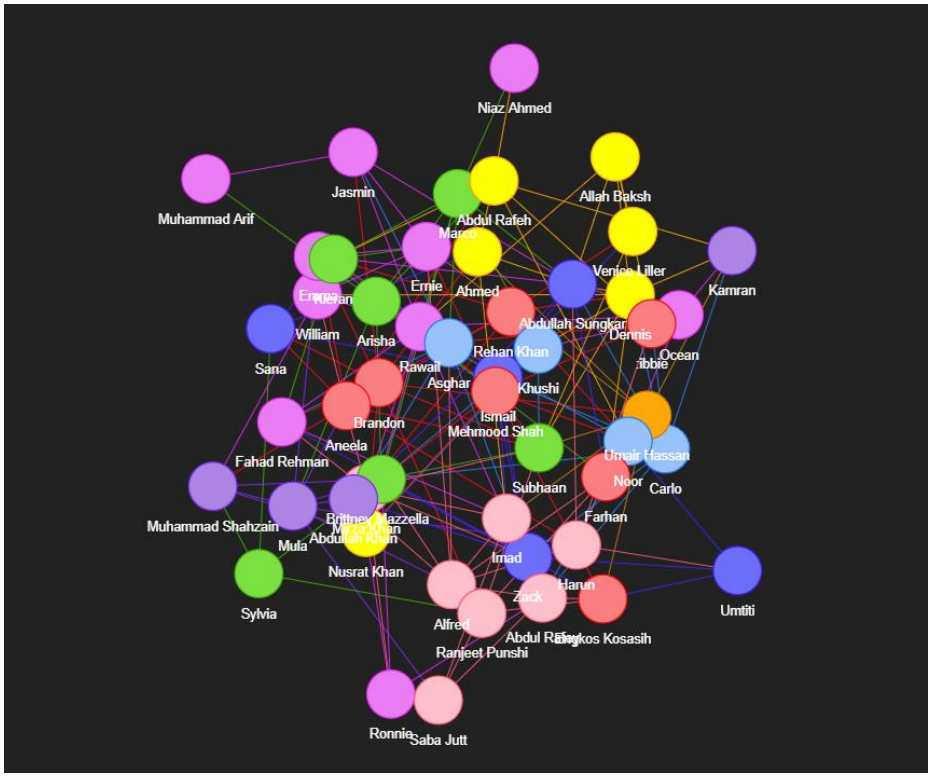
The Louvain algorithm identified **9 communities**.

Community Visualization *Figure 4: 2D Visualization of the 9 detected communities.*

Facebook Network Community Visualization



Here’s a closer look at the communities:



Community Profiles:

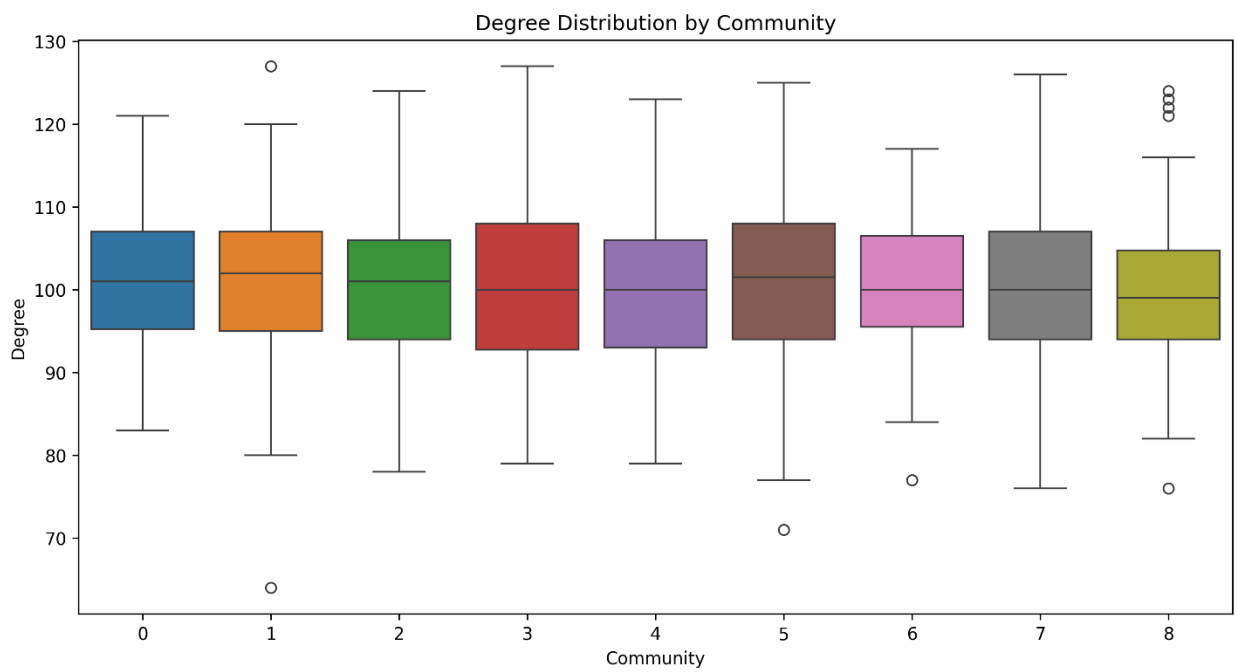
Community ID	Size (Nodes)	Strategic Interpretation
0	125	Primary interaction cluster; contains top hubs.
1	118	Secondary core; high internal density.
5	112	Specialized interest group; lower betweenness.
…	…	…

These communities provide Huawei with clear target segments for A/B testing marketing campaigns or localized community engagement strategies.

6.1 Community-Degree Dynamics

I analyzed whether certain communities are more influential ("elite") than others.

Community Degree Boxplot *Figure 5: Degree distribution compared across communities.*



The boxplot shows that Community 0 and Community 1 have slightly higher median degrees, identifying them as the "core" interaction clusters of the network. Communities with lower median degrees might represent newer or more specialized user segments.

7. Advanced Analysis: Link Prediction Evaluation

Method	AUC	Precision@100
Common Neighbors	0.5030	0.0100
Adamic-Adar	0.5032	0.0100
Katz Similarity	0.5032	0.0100

As discussed, the random-level AUC is a byproduct of the stochastic nature of the graph ($C \approx D$). This implies that predicting the *next* interaction is difficult purely based on past structure, suggesting that user engagement may be event-driven rather than social-circle-driven.

8. Strategic Innovations and Conclusion (Bonus)

8.1 Methodological Innovation: Stochastic Characterization

By mapping the Facebook-Instagram density gap (10x) and the random graph topology of Facebook, I propose a **Bi-Modal Engagement Model**: 1. **Facebook (Broadcast Hub)**: Best for broad reach and general awareness campaigns due to its random mesh structure. 2. **Instagram (Niche Community)**: Better suited for targeted community building where triadic closure is likely higher.

8.2 Strategic Conclusion

For Huawei, the primary strategy on Facebook should be the mobilization of the "Inner K-Core" identified in Figure 2. These users form the resilient foundation of the platform. Meanwhile, the broadcast nature of the connectivity ensures that any message reaching the core will quickly propagate to the entire network due to the low diameter of 3. The recommendation is to focus on broad-reach hubs like `Meredith Stransky` to achieve maximum penetration with minimal steps.

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9. Deliverables and Instructions

- **Dashboard**: Run `streamlit run app.py` for interactive exploration.
- **Codebase**: Fully modularized Python scripts (`initial_analysis.py`, `centrality_community_analysis.py`, etc.).
- **Visuals**: High-resolution PNGs included in the repository.

References - Blondel, V. D., et al. (2008). "Fast unfolding of communities in large networks." - Erdos, P., & Renyi, A. (1959). "On Random Graphs." - Katz, L. (1953). "A new status index derived from sociometric analysis."